# Recognising rare narratives in English news: lexical features, document embeddings and multilingual training

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**Abstract**

On English news, document-averaged F1 can obscure substantial differences between articles labelled only as Other and those annotated with specific narratives. Experiments on 399 English articles from SemEval-2025 Task 10 compare lexical features, frozen document embeddings, class weighting and multilingual training across the same evaluation folds. A balanced word-based support vector machine obtained the highest observed fine-grained document F1 (0.4617). Combining word features and embeddings scored 0.4572 overall, but raised F1 on articles with specific labels from 0.1276 to 0.2103 relative to words alone. Adding four languages to a fixed multilingual encoder recovered 29 of 191 rare-label annotations, compared with five under English-only training, while increasing false positives from 17 to 104 in this subset. The paired interval for the overall multilingual difference included zero. These results identify which annotations each intervention recovers and the precision cost of that recovery.

**Keywords**

narrative classification, multilabel learning, rare labels, multilingual transfer

## 1. Introduction and motivation

SemEval-2025 Task 10 asks how online news portrays events and actors in climate change and the Russia–Ukraine war. In its narrative classification subtask, a document can receive several labels at two levels of specificity [1]. An article may invoke a secondary narrative briefly, quote a source advancing it, or describe a position to contest it. The occurrence of relevant vocabulary is therefore an imperfect guide to the set of narratives that an annotator assigns.

Which modelling choices help recover specific subnarratives when English training examples are scarce? Three comparisons address distinct parts of this question. First, sparse word features and frozen embeddings offer different ways to recognise similar passages, and processing a whole article may preserve evidence beyond its opening. Second, increasing the loss assigned to minority examples may change which labels a classifier emits. Third, extra annotated articles in other languages may provide evidence for labels that occur only a handful of times in English. The contribution is a common evaluation of these choices, including document-level outcomes, label-frequency analyses and inspection of individual errors.

## 2. Related work

Piskorski et al. describe a multilingual, topic-specific taxonomy and distinguish assigning narratives from extracting textual evidence for a dominant narrative [1]. Here the prediction unit is the entire article. A useful classical baseline is a linear support vector machine (SVM) with sparse term features. Joachims showed why these classifiers suit text categorisation with many potentially relevant dimensions [2]. Sparse features also provide a straightforward way to compare the effect of class weighting while holding the representation fixed.

Reimers and Gurevych trained sentence encoders whose pooled representations support semantic similarity comparisons [3]. Their multilingual distillation work aligns embeddings across languages [4]. These approaches motivate using pretrained representations with a small supervised classifier when labelled English documents are limited. A news article, however, often exceeds a sentence encoder's

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input length. This motivates two comparisons: the opening block versus the full article, and English-only versus multilingual training with a shared encoder.

# 3. Proposed approach

## 3.1. Corpus and label inventory

The initial training corpus comprised 1,699 news articles. After overlap checks and exclusions, 1,688 articles remained: 399 English, 400 Bulgarian, 359 Hindi, 397 Portuguese and 133 Russian articles. The Russian subset comes from the main training release. A second release was excluded because its coverage differs and overlaps with the first. In the original training data, 23 groups contained identical normalised texts, and 17 of these groups had differing label sets. Each document retained its original annotations. To limit leakage, English documents from the same group were assigned to the same evaluation partition.

The label inventory used for evaluation contains 22 coarse labels and 96 fine labels, including the global *Other* label. A comparison of the released v3 list with the corpus annotations and the organisers' taxonomy identified labels missing from its 91 fine classes. The primary score uses the extended inventory so that those annotations count in evaluation. This choice matters when comparing numbers across studies. In the English training corpus, 31 fine-grained labels have one to four annotated articles, including eight with a single article; a further 12 labels are unobserved. An individual fold can therefore lack positive training examples even for a label required at evaluation.

## 3.2. Representations and classifiers

A constant baseline assigns *Other* to every article. For the lexical classifiers, the TF–IDF vocabulary and inverse document frequencies were estimated on the training partition of each fold. Word features used unigrams and bigrams, sublinear term frequency and a vocabulary cap of 15,000. Character features used within-word-boundary sequences of length three to five. Logistic regression and linear SVMs were evaluated with each of the three feature sets. A separate binary decision was learned per label. The balanced word SVM changed the class weights of each binary task while keeping its features and regularisation settings fixed.

The frozen `all-mpnet-base-v2` encoder provided English semantic features in two settings: the opening block alone and contiguous blocks covering the full article. Each block contained at most 384 tokens. The lexical–semantic system concatenated word TF–IDF and the full-article vector with equal weights and applied a final L2 normalisation.

A diagnostic on 16 training articles tested whether a linear head could fit frozen mBERT representations. It reached document-averaged F1 of .9792 at both label levels on that same subset; this measures fitting capacity rather than generalisation. A separate one-update audit on a fresh pretrained model recorded finite gradients and weight changes in the inspected encoder and classifier tensors. For the three-fold evaluation, frozen mBERT supplied a 768-dimensional pooled representation of each article, truncated to 512 tokens. A linear head jointly produced 22 coarse and 96 fine outputs and was trained with binary cross-entropy. The checkpoint and separate coarse/fine thresholds were selected using inner validation. At 512 tokens, about 77% of English articles exceeded the mBERT limit and 80% exceeded the XLM-R limit, motivating full-document aggregation.

## 3.3. Controlled multilingual comparison

Both conditions used the frozen `paraphrase-multilingual-mpnet-base-v2` checkpoint, which has an XLM-R backbone. Each article was split into non-overlapping blocks of up to 128 tokens, including special tokens. Within each block, mean pooling over non-padding hidden states yielded a vector that was then normalised. The document representation was the normalised weighted mean

of the block vectors, with weights proportional to their content-token counts. This produced one 768-dimensional vector per article from 8,977 blocks. The encoder used no task labels during extraction.

The classifiers were one-vs-rest linear SVMs with L2 regularisation, squared hinge loss, $C = 1$, tolerance $10^{-4}$ and a 10,000-iteration cap. In each fold the multilingual condition added the same 1,289 auxiliary articles to the English inner training set. Validation and outer evaluation remained English. Within each level and fold, a single threshold applied to all labels. Selection maximised inner-validation F1, with ties resolved in favour of the larger threshold. The grid ran from $-1.0$ to $0.3$ in steps of $0.1$, plus $0.5$. Scores strictly above threshold were selected; if none qualified, the highest-scoring label was returned. A label absent in binary training received a constant negative decision. Candidate rules making *Other* exclusive or adding a predicted child's parent showed no consistent validation gain, so the multilingual comparison decoded each level independently.

# 4. Evaluation

## 4.1. Protocol and metrics

The same 399 English articles were evaluated in three outer folds of 133, 134 and 132 documents. Their respective inner training sets contained 213, 212 and 214 English documents, each with 53 additional English validation documents. Each outer article contributed exactly one held-out prediction per system. Scores were calculated from the pooled held-out predictions. For article $i$ with gold labels $Y_i$ and predictions $P_i$, the primary metric is

$$F_{\mathrm{doc}} = \frac{1}{399} \sum_{i=1}^{399} \frac{2|Y_i \cap P_i|}{|Y_i| + |P_i|}.$$

It gives each article equal weight. Label-level precision and recall pool individual decisions across documents, so recovered annotations and false positives can be counted directly. Paired bootstrap intervals were estimated from 2,000 resamples of the 395 English document groups. Both systems were evaluated on each draw, with all documents in a group sampled together. These intervals are conditional on the fitted models and fixed splits; they do not capture training or split variability, fully account for dependence from overlapping cross-validation training sets, or correct for exploratory model selection.

**Table 1**
Document-averaged F1 on the 399 English outer-fold articles. Thresholds were selected on inner validation. The last two rows share one multilingual encoder.

| System | Coarse F1 | Fine F1 |
|---|---|---|
| Always Other | .4236 | .4236 |
| Word TF–IDF + logistic regression | .4731 | .4282 |
| Character TF–IDF + logistic regression | .4670 | .4252 |
| Word and character TF–IDF + logistic regression | .4782 | .4233 |
| Frozen mBERT + linear head | .4724 | .3929 |
| Word TF–IDF + SVM | .5403 | .4586 |
| Character TF–IDF + SVM | .5514 | .4446 |
| Word and character TF–IDF + SVM | .5543 | .4547 |
| English MPNet, first block + SVM | .5090 | .4305 |
| English MPNet, full article + SVM | .5434 | .4550 |
| Word TF–IDF and full English MPNet + SVM | **.5598** | .4572 |
| Word TF–IDF + balanced SVM | .5498 | **.4617** |
| Multilingual encoder + SVM, English training | .5417 | .4407 |
| Multilingual encoder + SVM, five-language training | .5272 | .4326 |

## 4.2. What the aggregate scores conceal

Word TF–IDF with an SVM outperformed logistic regression on fine labels. Full-article English MPNet raised fine F1 from .4305 to .4550 relative to using the first block. Combining full-article embeddings with words gave the highest coarse score (.5598). The word SVM and its lexical–semantic counterpart were close on fine F1 (.4586 and .4572), yet performed very differently on the two article types. On gold *Other*-only articles the respective scores were .9090 and .7934. On articles with specific labels they were .1276 and .2103. Frozen full-article MPNet alone scored .8031 and .1993 in the same two groups. All paired intervals for overall differences among these three systems included zero. Reporting the composition of errors helps explain why their aggregate scores converge.

Inspection of 12 selected articles revealed missed secondary narratives, confusion between neighbouring labels and false positives on reported or negated statements. These observations illustrate challenges in assigning a document-level label when an article quotes multiple voices. They do not estimate the prevalence of each error type.

Balancing the word SVM improved observed fine F1 by .0031, with a paired interval including zero. The average number of predicted fine labels rose from 1.90 to 2.31 per article. Micro recall increased from .3269 to .3737 while micro precision fell from .3773 to .3554. Among classes with five to nine English training examples, recall rose from 11.2% to 19.3%. For classes with one example, both systems recovered zero occurrences. The strongest aggregate score thus came with more predicted labels and a measurable precision cost.

## 4.3. When do auxiliary languages help?

The overall multilingual difference (five languages minus English only) was $-.0081$ fine F1, with paired 95% interval $[-.0443, .0280]$. The coarse difference was $-.0145$ $[-.0535, .0253]$. For fine labels, multilingual training gained .0308 and .0256 on the first two folds and lost .0814 on the third. Treating only the direction of each fold as evidence would obscure the size of the third-fold loss.

**Table 2**
Specific fine labels by support in the English inner training fold. Each pair gives English-only / multilingual results on the same held-out annotations. The global Other label is excluded. The last row is a subset of the 1–4 band.

| English support | Gold | Recovered | False positives | Recall (%) |
|---|---|---|---|---|
| 0 | 19 | 0 / 0 | 0 / 7 | 0.0 / 0.0 |
| 1–4 | 210 | 6 / 30 | 19 / 106 | 2.9 / 14.3 |
| 5–9 | 269 | 46 / 82 | 151 / 178 | 17.1 / 30.5 |
| 10 or more | 208 | 80 / 97 | 185 / 205 | 38.5 / 46.6 |
| 1–4 reaching 5+ with auxiliary data | 191 | 5 / 29 | 17 / 104 | 2.6 / 15.2 |

For the rare categories that reached at least five training examples after adding other languages, all 24 additional net recoveries in the 1–4 band arose in this subset (Table 2). The additional 87 false positives lowered its precision from 22.7% to 21.8%. The 5–9 band behaved differently: precision increased from 23.4% to 31.5% alongside recall. Additional examples can therefore increase both measures for some support ranges even when aggregate document F1 declines. These subgroup results are descriptive and depend on the chosen thresholds.

A fixed zero threshold provides a sensitivity check. Fine F1 was .4344 for English-only and .4106 for multilingual training, preserving their order. Coarse F1 changed to .5217 and .5285, reversing the ordering. In the multilingual fine classifier, 244 of 399 articles changed predictions relative to tuned thresholds. Repeating the English SVM with solver seeds 13, 42 and 87 at fixed folds and embeddings yielded identical predictions on all 399 articles at both levels. This check isolates the solver's contribution to variability.

## 5. Conclusions and limitations

For these English news articles, balancing a lexical classifier gave the highest observed overall fine F1. Full-document semantic features improved performance on specifically annotated articles, and auxiliary languages recovered more annotations in several rare-label bands. The intended use matters: a reader screening articles for a rare narrative may value additional recoveries, while the accompanying false positives increase the review workload. Reporting both counts exposes this trade-off.

Only 399 English documents contribute to outer evaluation, and scarce labels can be absent in a training fold. Some experiments were motivated by errors observed in earlier outer predictions, making the study exploratory. A separate set of 41 English test articles had already been examined during earlier development and therefore could not provide an independent final evaluation. Grouping reduces overlap from identical or lexically similar documents, but translations and paraphrases may escape detection. Multilingual training also changes data volume and label and domain frequencies together. Finally, seed checks cover the English SVM solver at fixed representations and folds. Future evaluation on independently collected English articles could test whether the gains for specific labels persist across topics and publishers.

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